

Mathematics Courses

Runqing Zhuo

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- Independent Reading
- Pseudodifferential Operators
- Riemannian Geometry

MATH 499

graduate level

graduate level

Course Details

Mathematical Analysis (I)

Instructor: Prof. Cheng, Chongqing Score: 93.1

Fall 2022, Nanjing University

Textbooks: Lecture notes (Chinese) by Prof. Cheng

Main Topics: Real number system (completeness), limits of sequences/series, differentiation (MVT, Taylor's expansion), the Riemann integral (FTC).

Higher Algebra (I)

Instructor: Prof. Ji, Qingzhong Score: 94.5

Fall 2022, Nanjing University

Textbooks: Liu, *Higher Algebra (Chinese)*

Main Topics: Polynomial ring theory, determinants, systems of linear equations, matrix theory.

Analytic Geometry

Instructor: Prof. Zhang, Gaoferi Score: 97.5

Fall 2022, Nanjing University

Introduction to Number Theory

Instructor: Prof. Qin, Hourong Score: 90

Fall 2022, Nanjing University

Mathematical Analysis (II)

Instructor: Prof. Cheng, Chongqing Score: 93

Spring 2023, Nanjing University

Textbooks: Lecture notes (Chinese) by Prof. Cheng

Main Topics: Uniform convergence, multivariable differentiation (Total derivative, Jacobian, inverse/implicit function Thms), multiple integrals, line/surface integrals (Green's/Gauss's Thms), intro to Fourier Analysis/Distributions.

Higher Algebra

Instructor: Prof. Ji, Qingzhong Score: 93

Spring 2023, Nanjing University

Textbooks: Liu, *Higher Algebra (Chinese)*

Main Topics: Vector spaces, linear maps, canonical forms (Jordan), inner product spaces (Euclidean/Unitary), quadratic forms, symplectic spaces.

Mathematical Analysis (III)

Instructor: Prof. Cheng, Chongqing Score: 94

Fall 2023, Nanjing University

Textbooks: Stein & Shakarchi, *Fourier analysis* ; Arnold, *Mathematical methods of classical mechanics*

Main Topics: Fourier series, L^2 theory, and the Fourier Transform on $\mathbb{R}^n$ (including Plancherel's Theorem and the Uncertainty Principle). The application of these tools, including Tempered Distributions, to solve fundamental problems in Partial Differential Equations (e.g., heat and wave equations); Intro mathematical methods of classical mechanics (Legendre Transform, Lagrangian, Hamiltonian).

Ordinary Differential Equations

Instructor: Prof. Cheng, Wei Score: 86

Fall 2023, Nanjing University

Textbooks: Lecture notes (Chinese) by Prof. Cheng; Walter, *Ordinary Differential Equations*

Main Topics: Existence/uniqueness theory (Picard-Lindelöf, Peano), continuous dependence, stability theory, Gronwall's inequality, the Method of Characteristics for first-order PDEs.

Abstract Algebra

Instructor: Prof. Chen, Ke Score: 91.6

Fall 2023, Nanjing University

Textbooks: Lecture notes (Chinese) by Prof. Chen

Main Topics: Group theory (including Sylow Theorems), ring theory (ideals, PIDs, quotients, polynomial rings), field extensions, splitting fields, Galois Theory.

Real Analysis

Instructor: Prof. Lv, Yong Score: 90

Fall 2023, Nanjing University

Textbooks: Lecture notes (Chinese) by Prof. Cheng; and I read Stein & Shakarchi, *Real analysis* (whole book) myself

Main Topics: Lebesgue Measure Theory (construction, measurable functions, integration), convergence theorems (MCT, DCT), differentiation theory (BV/AC functions, Lebesgue Differentiation Thm), and L^p spaces.

Functions of one Complex Variable

Instructor: Prof. Liao, Liangwen Score: 90

Spring 2024, Nanjing University

Textbooks: I read Stein & Shakarchi, *Complex analysis* (Ch.1,2,3,4,5,8) myself

Main Topics: Holomorphic/harmonic functions (Cauchy theorem/formula, singularities, Maximum modulus principle), meromorphic functions and the Residue theorem, and Conformal mappings (Schwarz lemma, Riemann mapping theorem).

Foundations of Probability Theory

Instructor: Prof. Dai, Xiongping Score: 94

Spring 2024, Nanjing University

Textbooks: Li, *Foundations of Probability Theory* (Chinese)

Main Topics: Axiomatic and geometric probability, conditional probability, and statistical independence (Bayes', Bernoulli trials). Introduces random variables, vectors, and their distributions. Focuses on numerical characteristics: expectation, variance, and characteristic functions. Concludes with convergence types, the strong and weak Laws of Large Numbers, and the Central Limit Theorem.

Numerical analysis

Instructor: Prof. Deng, Weibing Score: 90.5

Spring 2024, Nanjing University

Textbooks: Lin, *Numerical Computation Methods* (Chinese)

Main Topics: Numerical solutions for ODEs (One-step and Linear Multistep methods), direct/iterative methods for linear systems, numerical integration/differentiation (Newton-Cotes) and orthogonal polynomials.

Selected Topics in Analysis

Instructor: Prof. Qiu, Hua Score: 95

Spring 2024, Nanjing University

Textbooks: Heinonen, *Lectures on Analysis on Metric Space* (Chap.1-4); Falconer, *Fractal Geometry*

Main Topics: Analysis on Metric Spaces: Hausdorff measure/dimension, covering lemmas, singular integrals and geometric measure theory, the Hardy-Littlewood maximal operator, and Poincaré inequalities.

Theoretical Mechanics

Instructor: Prof. Fan, Renhao Score: 100

Spring 2024, Nanjing University

Textbooks: I read Arnold, *Mathematical methods of classical mechanics* myself

Main Topics: Foundational principles of classical mechanics: Lagrangian mechanics (Euler-Lagrange equations) and Hamiltonian mechanics.

Mathematical Methods for Physics

Instructor: Prof. Wu, Shengjun Score: 93

Spring 2024, Nanjing University

Textbooks: Liang, *Mathematical Methods for Physics* (Chinese)

Main Topics: Application-focused mathematical techniques: Complex Analysis and its applications, the variational method, Green's function and integral transform, theory of linear ODEs and methods for PDE series solutions (e.g., Fourier series).

Partial Differential Equations

Instructor: Prof. Ruan, Zhuoping Score: 93

Fall 2024, Nanjing University

Textbooks: Jiang, *Lecture Notes on Equations of Mathematical Physics* (Chinese)

Main Topics: Derivation of PDEs from conservation laws (momentum, energy, mass) and variational principles. Solution methods for Wave, Heat, and Potential equations: characteristics, Fourier transform, separation of variables, Green's functions, and maximum principles. Includes well-posedness, energy estimates, and an introduction to linear PDE classification.

Topology

Instructor: Prof. Shi, Weixue Score: 93

Fall 2024, Nanjing University

Textbooks: You, *Lectures on Basic Topology* (Chinese)

Main Topics: Point-Set Topology (axioms, compactness, connectedness, Urysohn lemma, Tietze extension theorem), metric spaces and intro to Algebraic Topology (homotopy, the fundamental group).

Galois Theory

Instructor: Prof. Qin, Hourong Score: 97

Fall 2024, Nanjing University

Textbooks: N/A

Main Topics: Advanced seminar covering the fundamental concepts of Galois Theory: field extensions, the Galois group, and the Fundamental Theorem. Explores applications like solvability by radicals and classical construction problems. Seminar topics include general equations, cyclotomic and Kummer extensions, infinite Galois theory, and group cohomology.

Analysis

Instructor: Prof. Sun, Yongzhong Score: 88

Fall 2024, Nanjing University

Textbooks: Lecture notes (Chinese) by Prof. Sun; Folland, *Real Analysis* (Ch.1-7)

Main Topics: Measure Theory (Hahn-Kolmogorov, Fubini/Tonelli, Radon-Nikodym), the basics of Functional Analysis (Bounded Linear Operators, Dual Space), and Riesz Representation Thm, weak/weak* topologies, Banach-Alaoglu).

Harmonic Analysis

Instructor: Prof. Ruan, Zhuoping Score: 92

Fall 2024, Nanjing University

Textbooks: Grafakos, *GTM249 (chapter 1,2,5,6) & GTM 250 (chapter 3)*

Main Topics: Fundamentals of real analysis, including operator interpolation, Fourier transforms, convolutions, and approximate identities. Hardy-Littlewood maximal operator and Riesz potentials/Hardy-Littlewood-Sobolev inequality. Singular integral operators of convolution type, Littlewood-Paley theory, multiplier theory, and the function spaces H^1 and BMO.

Functional Analysis

Instructor: Prof. Qin, Lizhen Score: 97

Spring 2025, Nanjing University

Textbooks: Lecture notes (Chinese) by Prof. Qin

Main Topics: Covers metric spaces (completeness, compactness, fixed point theorems), and core Banach and Hilbert space theory (bases, orthogonal systems). Focuses on bounded linear operators: fundamental theorems (Open Mapping, Closed Graph, Uniform Boundedness), dual spaces, adjoint operators, and spectral theory, including self-adjoint and compact operators.

Differential Geometry

Instructor: Prof. Chen, Xuezhong Score: 92

Spring 2025, Nanjing University

Textbooks: Do Carmo, *Differential Geometry of Curves & Surfaces*

Main Topics: Classical Differential Geometry: Frenet-Serret apparatus (curvature/torsion), surface theory via the first/second fundamental forms, Gauss map, and Gaussian/Mean curvature for surfaces in $\mathbb{R}^3$.

Analysis (II)

Instructor: Prof. Sun, Yongzhong Score: 90

Spring 2025, Nanjing University

Textbooks: Xia, *A Second Course in Functional Analysis* (Chinese)

Main Topics: Analysis in Banach Spaces: Geometric foundations (Hahn-Banach Theorem/Convex set separation) and the calculus of vector-valued functions (including Bochner integral).

Manifold and Geometry

Instructor: Prof. Xu, Yiyang Score: 93

Spring 2025, Nanjing University

Textbooks: Mei, *Manifold and Geometry*; I read Lee, *Introduction to Smooth Manifolds* (Ch.1-10) myself

Main Topics: Differentiable Manifolds: charts/atlas, smooth maps, tangent and cotangent bundles, partitions of unity, the tangent/cotangent bundles, vector fields/flows, Lie bracket, derivatives (differentials), and the Regular Value Theorem (submanifolds).

From calculus to cohomology

Instructor: Prof. Shi, Yalong Score: 100

Spring 2025, Nanjing University

Textbooks: Lecture notes by Prof. Shi(Chinese); Madsen, *From calculus to cohomology*; Bott & Tu, *Differential Forms in Algebraic Topology* (Ch. 1-5)

Main Topics: Differential forms, exterior algebra, chain complexes, and de Rham cohomology. Differentiable manifolds, differential forms on manifolds, integration, and computational methods for de Rham cohomology. Concepts of mapping degree and vector field index, leading to the Poincaré-Hopf and Poincaré Duality theorems. Advanced topics include vector bundles, connections, curvature, and characteristic classes (Euler and Chern classes).

Sino-Italian mathematical course in analysis

Instructor: Prof. Sorrentino, Alfonso; Prof. Cannarsa, Piermarco Score: 90 Spring 2025, Nanjing University

Course 1: Geometric Hamiltonian Dynamics

Instructor: Prof. Sorrentino, Alfonso

Textbooks: Moser & Zehnder, *Notes on Dynamical Systems*

Main Topics: Symplectic manifolds, Hamiltonian flows, action-angle variables, Poisson brackets, canonical transformations, and Arnold-Liouville theorem.

Course 2: Evolution Equations

Instructor: Prof. Cannarsa, Piermarco

Textbooks: Lecture notes by Prof. Cannarsa

Main Topics: the generation of strongly continuous semigroups of bounded linear operators, the solvability of the associated inhomogeneous Cauchy problem, and the basic controllability properties (namely, approximate, null, and exact controllability) of abstract linear control systems in continuous time.

MATH 656, Introduction to Partial Differential Equations

Instructor: Prof. Sijue Wu Score: 4.0

Fall 2025, University of Michigan

Textbooks: Chapters 1, 2, 3.6, 4, 5, 7 of F. John, and/or Chapters 1–3, 5 of Evans.

Main Topics: Nonlinear first order equations; linear elliptic, hyperbolic and parabolic equations; method of characteristics; energy method; maximum principle; Fourier transform; distributions; Sobolev spaces.

MATH 399, Independent Reading

Instructor: Prof. Tian Jing Score: 4.0

Fall 2025, University of Michigan

Textbooks: Journé, *Calderon-Zygmund Operators, Pseudo-Differential Operators and the Cauchy Integral of Calderon*, Chapters 0–3.

Main Topics: Preliminaries on $L^p_B(\mathbb{R}^n, d\mu)$; Hardy–Littlewood maximal operator; A_p weights; $BMO(\mathbb{R}^n, dx)$. I present two hours every week.

MATH 499, Independent Reading

Instructor: Prof. Xiao Ma Score: 4.0

Fall 2025, University of Michigan

Textbooks: Evans, *Partial Differential Equations*, Chapter 6; Reed & Simon, *Methods of Modern Mathematical Physics I*; Chen, *Pseudo-differential Operators*; Tao's blog; 247B notes.

Main Topics: Topics including elliptic PDE, functional analysis, recapping basic theory in harmonic analysis, pseudo-differential operators, Fourier restriction theory and modern harmonic analysis.

Pseudodifferential Operators

Instructor: Prof. Albert Ai Score: N/A

Spring 2026, BICMR

Textbooks: Bahouri et al., *Fourier Analysis and Nonlinear PDEs*; Taylor, *Pseudodifferential Operators and Nonlinear PDE*; Métivier, *Para-differential calculus*.

Main Topics: Introduces pseudodifferential and paradifferential operators with applications in nonlinear PDEs. Topics include Sobolev and Besov spaces, Littlewood-Paley decomposition, paraproducts, symbol classes, symbolic calculus, and nonlinear hyperbolic systems. (Prerequisites: Real and Fourier analysis.)

Riemannian Geometry

Instructor: Prof. Ge, Huabin Score: N/A

Spring 2026, BICMR

Textbooks: Do Carmo, *Riemannian geometry*; Petersen, *Riemannian geometry*; Wu et al., *Introduction to Riemannian Geometry* (Chinese).

Main Topics: Fundamentals of Riemannian Geometry: metric, connection, covariant derivative, geodesics, curvature, Jacobi fields, and variation formulas. Covers comparison theorems, the Morse Index Theorem, and the Sphere Theorem. Also includes an overview of advanced topics like Gromov-Hausdorff convergence and Ricci flow. (Prerequisites: Basic topology, ODEs, and differentiable manifolds.)